

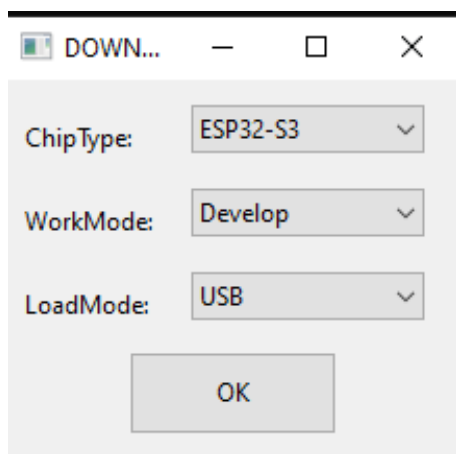
ESP32-S3: FIRMWARE BACKUP, RESTORE AND RECOVERY

Before downloading new firmware to the ESP32-S3, it is recommended that a full Flash backup is performed. This should allow the device to be restored to the original state.

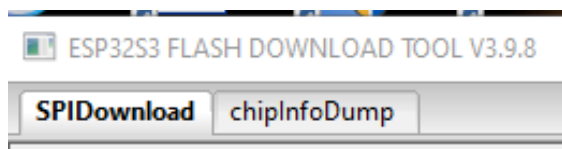
This process is at users own risk.

BACKUP

- Run the Espressif flash download tool
- Select ESP32-S3, Develop, USB and then press “OK”.

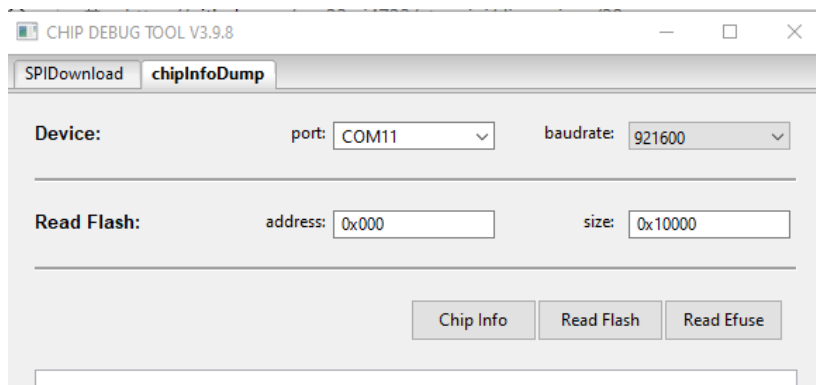


- Select the “chipinfoDump” tab



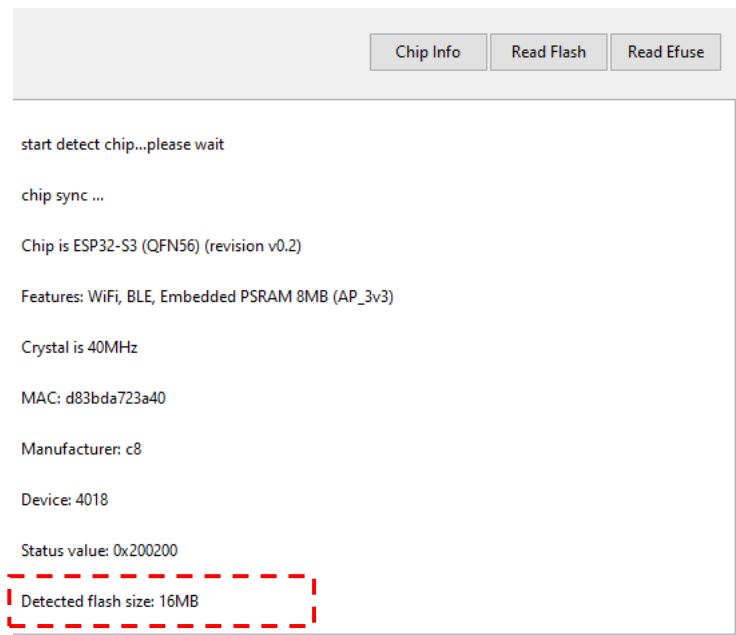
Set “port:” COM11 (Example)

Set “baudrate:” 921600



Select the “Chip info” tab

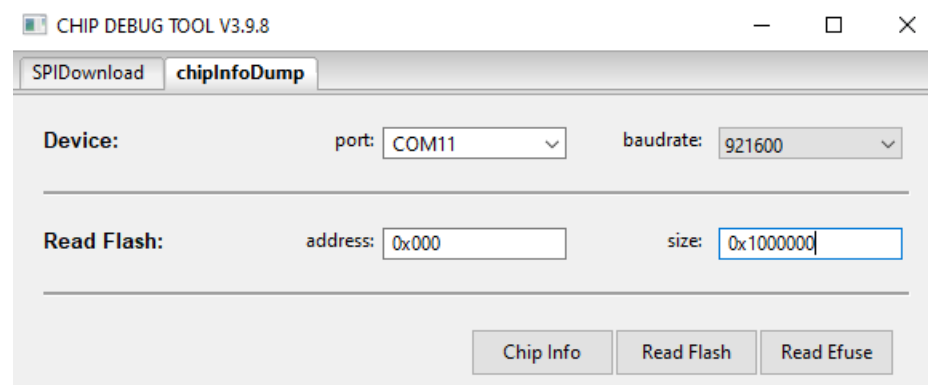
This will show the “Detected flash size”, in this example 16MB



Depending on the Flash size, set the “address” and “size” as shown in the table below.

Flash Size	address:	size:	Comments
8MB	0x000	0x800000	ESP32-S3 (N8R2)
16MB	0x000	0x1000000	ESP32-S3 (N16R8)

Check that all information is correct and then press “Read Flash”.

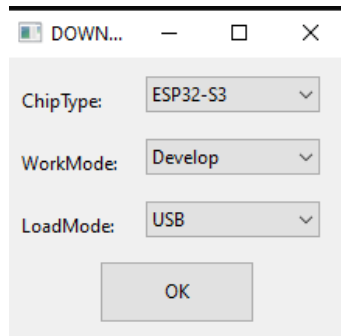


This will take several minutes to complete. Once it completed, the binary file will automatically be saved to the flash_dump directory. The file can be renamed if necessary.

Example: d83bda723a40_0x000_0x1000000_20250331_150943.bin

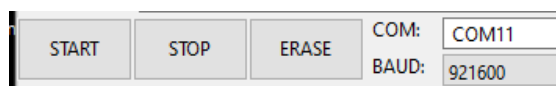
RESTORE

- Run the Espressif flash download tool
- Select ESP32-S3, Develop, USB



- The “SPIDownload” screen should open.

Set “COM” COM11 (Example)
Set “BAUD” 921600



Set the start address to 0x0.

Select the associated backup file for the device.



Check that all information is correct and then press "START".

It will take several minutes for the process to complete, wait for the "flash finish".

```
test offset : 0 0x0
case ok

Uploading stub...
Running stub...
Stub running...
Changing baud rate to 921600
Changed.
NO XMC flash detected!
SPI_BOOT_CRYPT_CNT 0
SECURE_BOOT_EN False
Compressed 16777216 bytes to 643392...

is stub and send flash finish
```



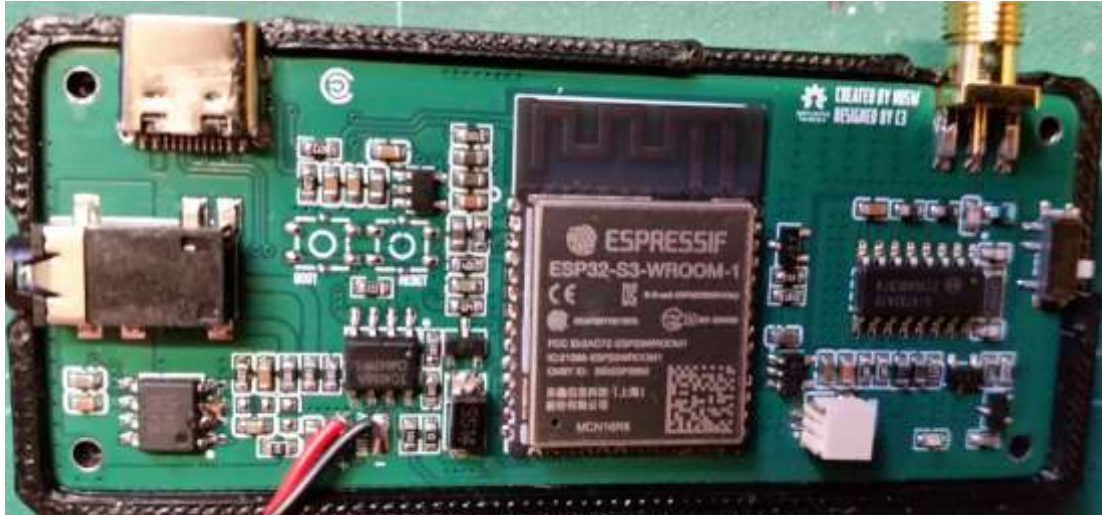
Cycle the power to the device, and it should boot with the original backed up firmware.

RECOVERY

If the flashing process fails it should be possible to put the ESP32-S3 into ROM boot mode.

Ideally, the receiver needs to have the “BOOT” and “RESET” buttons fitted to the PCB.

The positions on the PCB are shown below.



Max Arnold (R9UCL) GitHub page explains the recovery process in more detail, and it also covers backup and restore.

<https://esp32-si4732.github.io/ats-mini/recovery.html>

<https://esp32-si4732.github.io/ats-mini/mods.html>

Recovery

Please ensure the receiver has [BOOT](#) and [RESET](#) buttons on the PCB.

The recovery process is as follows:

1. Power on the receiver
2. Connect it via USB to a computer
3. Press and hold the BOOT button
4. Press the RESET button
5. Release the BOOT button

The receiver will enter into the recovery mode and you can flash the ESP32-S3 controller as usual.